

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Constraint-Based Problem Solving

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO5– Naturalization (BL5,BL6)	Assessment:	Assessment Tool 1 – Constraint Based Problem Solving
Weightage:	30% of CIA	Total Marks:	100
CO5 Statement:	<i>Analyze computational problems using backtracking techniques and complexity classes such as P, NP, NP-Hard, and NP-Complete for combinatorial problem solving.</i>		
Student Name:	Dinesh Karthick R	Reg. No.:	192524121
Section / Batch:	Batch 02	Date:	31-08-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Apply backtracking techniques systematically for combinatorial problem solving.
- Clearly classify problems under P, NP, NP-Hard, and NP-Complete categories wherever required.
- Provide proper justification, state-space representation, and complexity analysis for all solutions.
- Neat presentation and logical explanation are mandatory.

Question Type	Focus Area	Questions
Constraint Based Problem Solving	Analyze combinatorial problems using backtracking and complexity classes such as P, NP, NP-Hard, and NP-Complete.	Q1

SECTION A – CONSTRAINT-BASED PROBLEM SOLVING

Q1. Exam Timetabling (Graph Coloring / NP-Complete Problem) — 100 Marks

A university needs to schedule exams so that no student has overlapping exams while efficiently utilizing limited rooms. Clearly define constraints such as time slots, room capacity, and student conflicts. Analyze and prioritize these constraints using graph coloring concepts. Develop a solution using backtracking or CSP techniques. Apply appropriate methods to ensure conflict-free scheduling. Justify the correctness and feasibility of your approach.

Code of Conduct

I Dinesh Karthick R (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%	Clearly defines the problem and identifies all relevant constraints accurately	Identifies most constraints with minor gaps	Partial understanding; misses key constraints	Fails to identify constraints correctly
Constraint Analysis	25%	Analyzes constraints effectively and prioritizes them logically	Provides reasonable analysis with some prioritization	Limited analysis; weak prioritization	No meaningful analysis or prioritization
Solution Development	25%	Develops optimal and feasible solutions satisfying all constraints	Develops workable solutions with minor limitations	Solutions partially satisfy constraints	Solutions do not meet constraints
Application of Concepts	15%	Effectively applies appropriate concepts, methods, or tools	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply relevant concepts
Justification	10%	Provides strong justification for decisions with logical reasoning	Provides justification with some clarity	Weak or unclear justification	No justification provided

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	25%				
Constraint Analysis	25%				
Solution Development	25%				
Application of Concepts	15%				
Justification	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name:	Name:	Name:
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Output Screenshot — SIMATS Online Programming Lab



SIMATS Online Programming Lab
DAA PROGRAMMING

Dinesh Karthick R
192524121RA

ASSESSMENT- CO5-AT1-Constraint-Based Problem Solving

Language: Python C C++ Java

QUESTIONS

Q1
Exam Timetabling (Graph Coloring / NP-Complete Problem)
100 marks

1/1 answered

Q1 Exam Timetabling (Graph Coloring / NP-Complete Problem) 100 marks

A university needs to schedule exams so that no student has overlapping exams while efficiently utilizing limited rooms. Clearly define constraints such as time slots, room capacity, and student conflicts. Analyze and prioritize these constraints using graph coloring concepts. Develop a solution using backtracking or CSP techniques. Apply appropriate methods to ensure conflict-free scheduling. Justify the correctness and feasibility of your approach.

Your Code

```
1 #include <stdio.h>
2
3 int min(int a, int b, int c)
4 {
5     if (a < b && a < c)
6         return a;
7     else if (b < c)
```

Input

4
3
2
60
...

Output